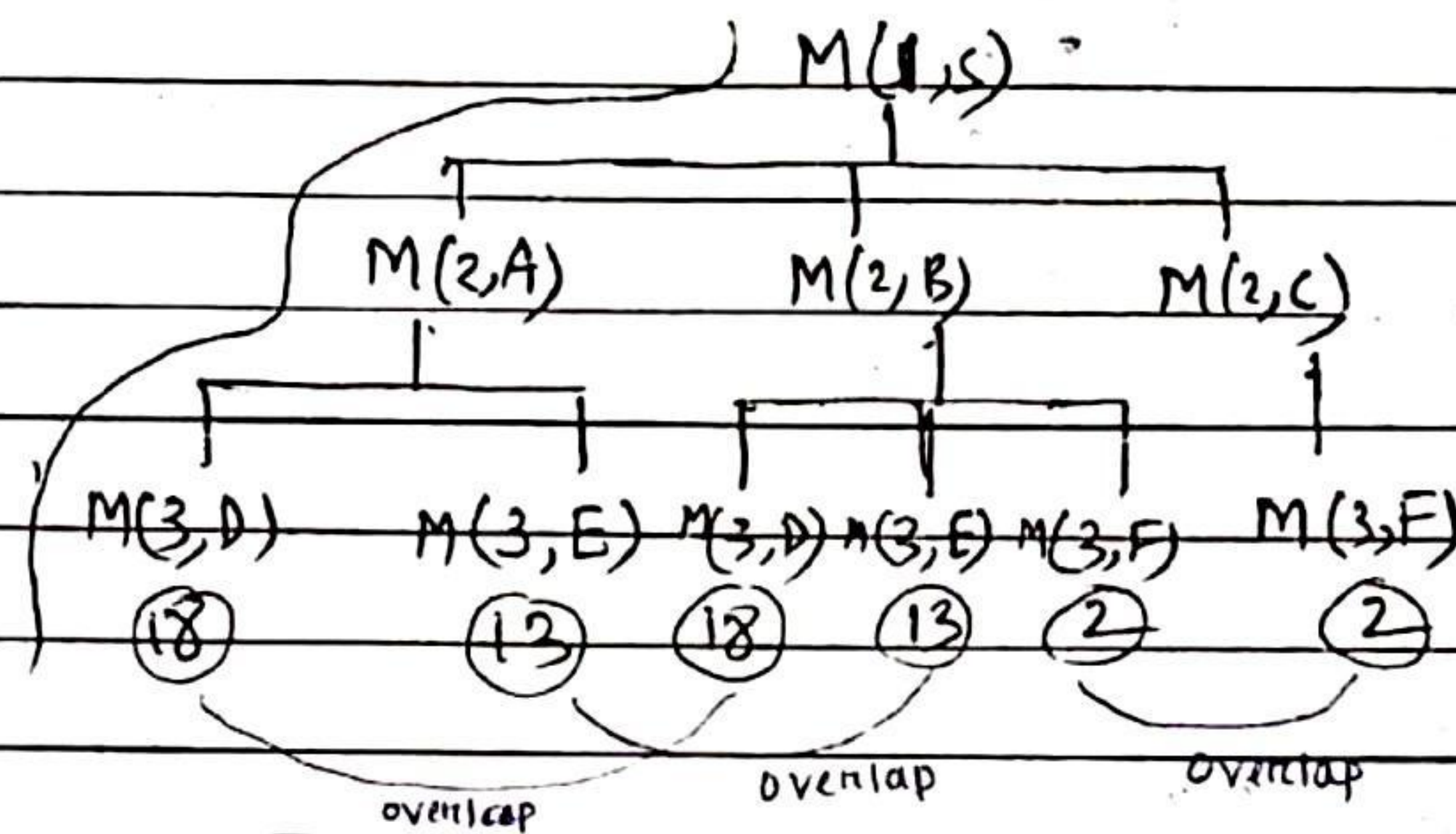


$$M(i, s) = \min \begin{cases} s \rightarrow A + M(2, A) \\ s \rightarrow B + M(2, B) \\ s \rightarrow C + M(2, C) \end{cases}$$

recursive tree -



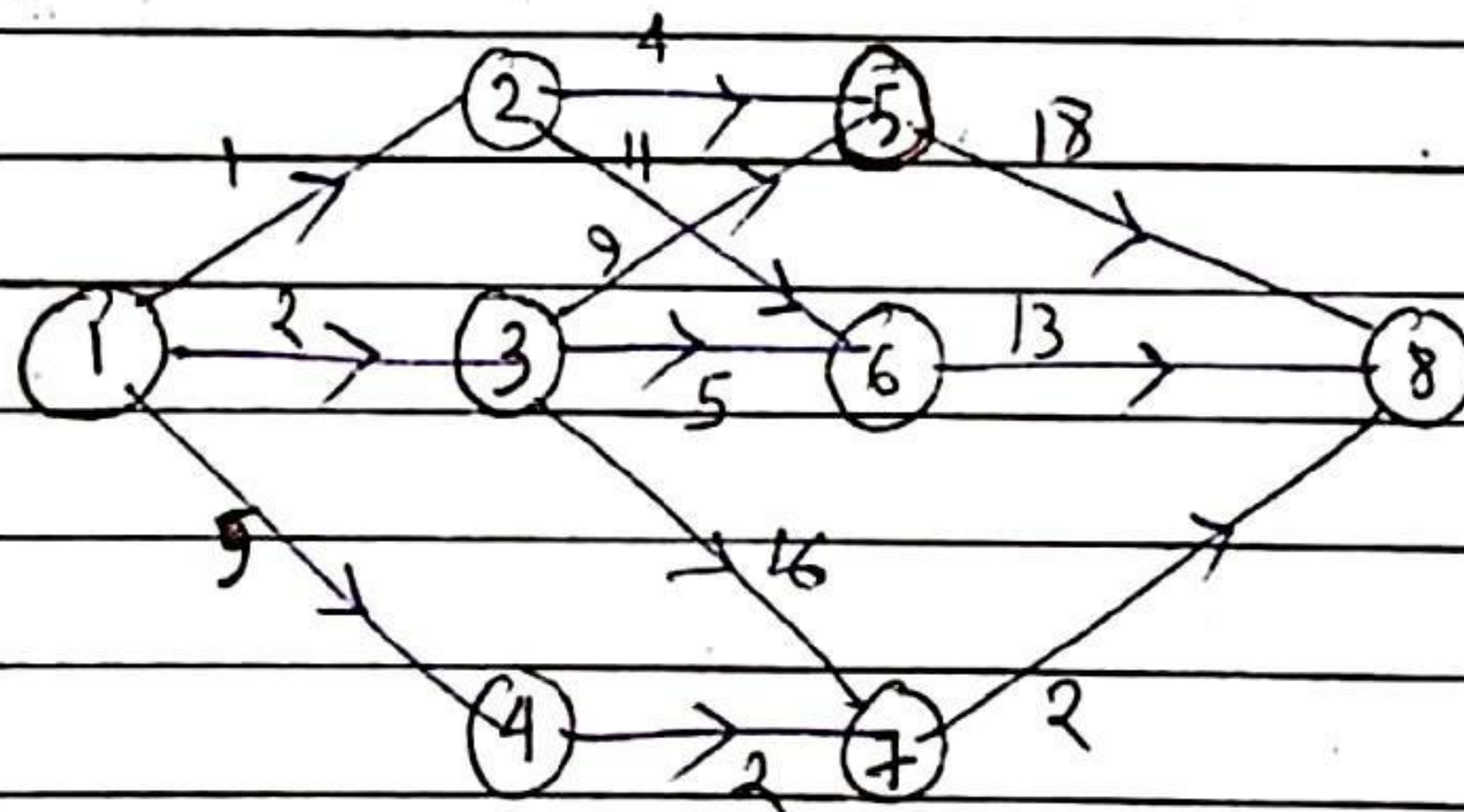
$M(N-1, i) = i \rightarrow T$ — Stop

no. level = K

space complexity = $O(K)$

time complexity = $O(K^n)$ — (exponential time)

- Apply Bottom up DP algo to ~~reduce~~ reduce time complexity by reduce overlapping —



no. of sub-problem = no. of vertices.

(array)	1	2	3	4	5	6	7	8
T =	9	22	18	4	18	13	2	0
	(7)	(6)	(5)	(4)	(3)	(2)	(1)	(1)

$$T[P] = \min_{\substack{\text{for all} \\ j = (i+1) \text{ to } n}} \left\{ \text{cost}(i, j) + T[j] \right\}$$

$$T[6] = \begin{cases} \text{cost}(6,7) + T[7] \\ \text{cost}(6,8) + T[8] \end{cases}$$

13 0

$$= 13$$

$$T[5] = \begin{cases} \text{cost}(5,6) + T[6] \\ \text{cost}(5,7) + T[7] \\ \text{cost}(5,8) + T[8] \end{cases}$$

18 0

$$= 18$$

$$T[1] = \begin{cases} \text{cost}(1,2) + T[2] = 23 \\ \text{cost}(1,3) + T[3] = 20 \\ \text{cost}(1,4) + T[4] = 9 \end{cases}$$

$$T[4] = \begin{cases} \text{cost}(4,7) + T[7] \\ \text{cost}(4,8) + T[8] \end{cases}$$

2 2

$$= 4$$

$$T[3] = \begin{cases} \text{cost}(3,5) + T[5] \\ \text{cost}(3,6) + T[6] \\ \text{cost}(3,7) + T[7] \end{cases}$$

9 18 = 27
5 13 = 18
16 2 = 18

$$T[2] = \begin{cases} \text{cost}(2,5) + T[5] \\ \text{cost}(2,6) + T[6] \end{cases}$$

4 18 = 22
11 13 = 24

$$O(n) \times O(n-1) = O(n^2)$$

Time complexity = no. of subproblem * time take to each problem.

$$= O(n)$$

$$= 1 + 2 + 3 + \dots + n$$

$$= 1 + 2 + 3 + \dots + (n-1)$$

$$= \frac{n(n+1)}{2}$$

$$= O(n^2)$$

$$= O(V^2)$$

$$\boxed{\text{Time complexity} = O(E)}$$